

Evolution of CNNs: From LeNet to ResNet

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1. Introduction

Have you ever wondered how your smartphone can recognise your face? Or how Google Photos automatically groups pictures of the same person together? The answer lies in a powerful technology called Convolutional Neural Networks (CNNs).

CNNs are a special type of Artificial Neural Network that are incredibly good at understanding images. Over the past few decades, CNNs have evolved dramatically from simple networks that could barely recognise handwritten digits, to super-deep networks that outperform humans at image recognition tasks.

In this blog, we take a journey through the history of CNNs from the humble LeNet in 1998 all the way to the powerful ResNet in 2015. By the end, you will understand how and why CNNs have become the backbone of modern computer vision.

What You Will Learn
What is a Convolutional Neural Network (CNN)?
How images are processed by CNNs layer by layer
The story of major CNN architectures: LeNet, AlexNet, VGGNet, GoogLeNet, ResNet
Why each new architecture was better than the previous one
A visual comparison of all major CNN models
Real-world applications of CNNs today

2. What is a Convolutional Neural Network (CNN)?

A CNN is a type of deep learning model specially designed to process structured grid data like images. Unlike a regular neural network that flattens an image into a long list of numbers, a CNN preserves the 2D structure of the image. This is the key reason CNNs are so powerful for image tasks.

2.1 How Does CNN Work? Layer by Layer

A CNN processes an image through several layers. Each layer learns different features starting from simple edges and corners, up to complex shapes and objects. The diagram below shows the full flow:

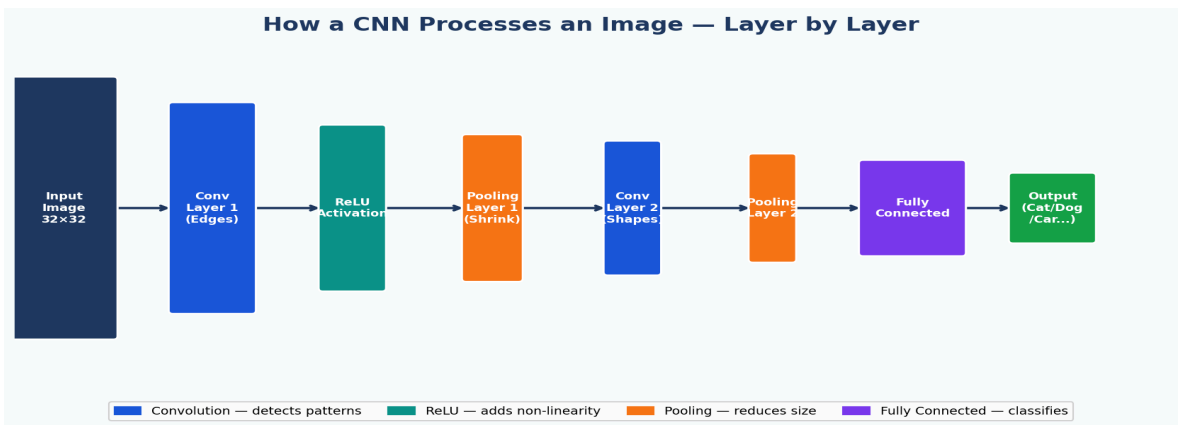


Figure 1: How a CNN processes an image from input to final output each layer learns more complex features

2.2 The Three Key Building Blocks

Layer	What It Does	Real-World Analogy
Convolutional Layer	Scans the image with a small filter to detect features like edges	Like a magnifying glass sliding over the image
Pooling Layer	Shrinks the feature map to reduce computation	Like zooming out to see the bigger picture
Fully Connected Layer	Make the final prediction (cat, dog, car...)	Like a judge giving a final verdict

3. CNN Evolution Timeline (1998–2015)

The story of CNNs is a story of solving one problem after another. Each new architecture fixed the weaknesses of the previous one. The timeline below shows the key milestones:

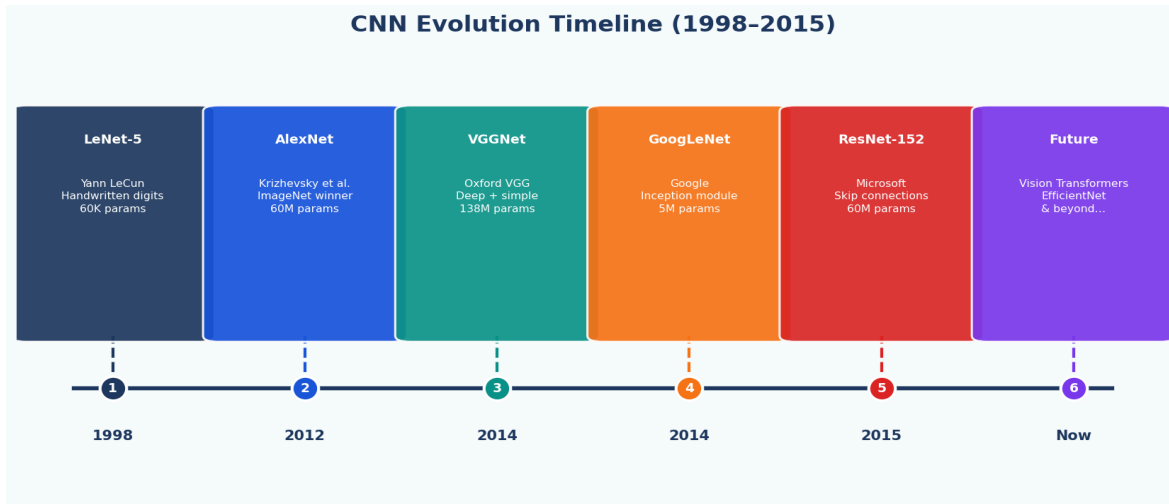


Figure 2: CNN evolution milestones each model introduced a key innovation that pushed the field forward

4. LeNet-5 (1998) The Pioneer

4.1 Background

LeNet-5 was created by Yann LeCun and his team at Bell Labs in 1998. It was designed to recognise handwritten digits specifically for reading ZIP codes on postal mail. This was a revolutionary idea at the time!

4.2 Architecture

LeNet-5 has 7 layers and only 60,000 parameters, tiny by modern standards, but groundbreaking in 1998. Here is the full architecture:

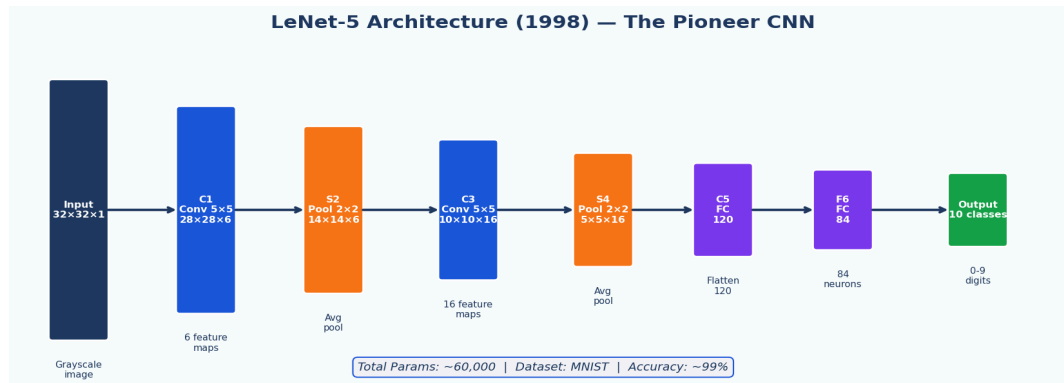


Figure 3: LeNet-5 architecture input image flows through Conv and Pooling layers before classification

4.3 Key Features

- Used Sigmoid and Tanh activation functions
- Used Average Pooling to reduce spatial dimensions
- Only 7 layers total very small by today's standards
- Trained on MNIST dataset (60,000 handwritten digit images)
- Achieved ~99% accuracy on digit recognition

Limitation of LeNet-5

LeNet-5 was great for simple grayscale digit images, but it could NOT handle:

- Colour images (RGB) it only accepted 32×32 grayscale input
- Complex real-world objects like cars, animals, and faces
- Large datasets MNIST was small; real-world data is much bigger

The world needed something much more powerful for the next challenge...

5. AlexNet (2012) The Deep Learning Revolution

5.1 Background

For over a decade after LeNet, CNNs were mostly ignored by the research community. Then in 2012, something incredible happened. Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton built AlexNet and entered the ImageNet Large Scale Visual Recognition Challenge (ILSVRC).

AlexNet won by a massive margin achieving a 15.3% error rate vs the runner-up's 26.2%. This was the moment the world realized deep learning was something special. It sparked the modern AI revolution we live in today!

5.2 Architecture

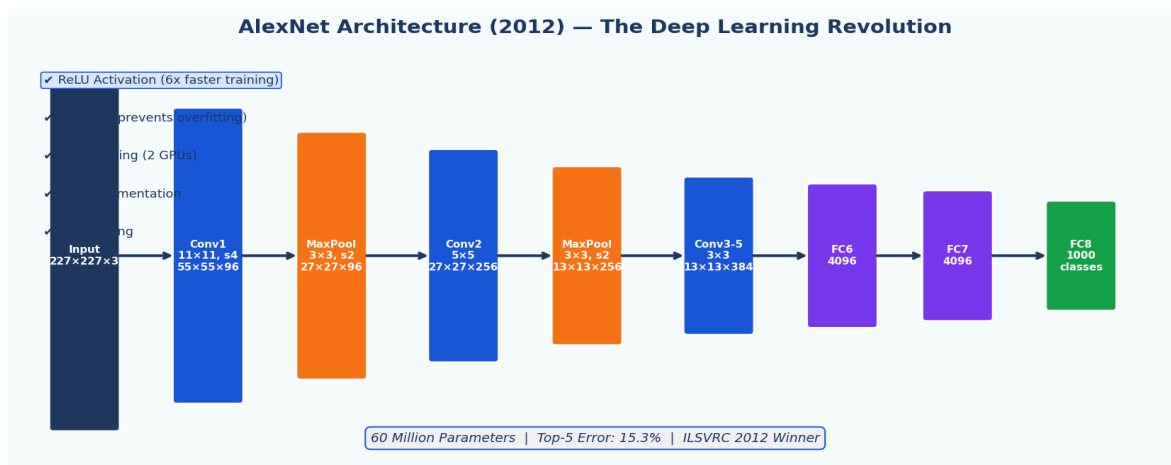


Figure 4: AlexNet architecture 8 layers, 60 million parameters, trained on two GPUs simultaneously

5.3 Why Was ReLU a Game Changer?

Before AlexNet, networks used Sigmoid or Tanh. These caused the Vanishing Gradient Problem where gradients become so tiny that the network stops learning. AlexNet introduced ReLU which solved this completely:

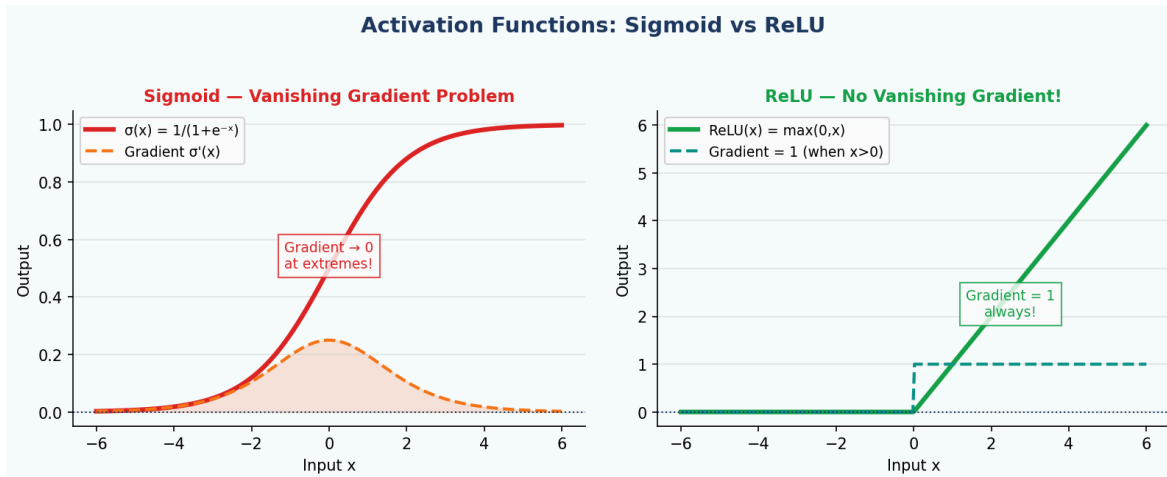


Figure 5: Sigmoid vs ReLU ReLU's gradient stays at 1 (for positive inputs), enabling much faster and deeper training

5.4 AlexNet's Key Innovations

Innovation	What It Does	Why It Mattered
ReLU Activation	$f(x) = \max(0, x)$	6× faster training than Sigmoid
Dropout (50%)	Randomly turns off neurons	Prevents overfitting
Data Augmentation	Flipping, cropping images	More training data from same images
GPU Training	Trained on 2 NVIDIA GPUs	Made deep training practical
Max Pooling	Takes max value in region	Better than Average Pooling

6. VGGNet (2014) Going Deeper with Small Filters

6.1 Background

In 2014, Karen Simonyan and Andrew Zisserman from Oxford's Visual Geometry Group (VGG) asked a simple but powerful question: What if we just use very small filters (3×3) and make the network much deeper?

VGGNet's Core Insight
Instead of one large 7×7 filter, use THREE stacked 3×3 filters.
Why is this better?
• Three 3×3 filters see the same receptive field as one 7×7 filter
• But they have FEWER parameters (27 vs 49 weights per channel)
• And they have MORE non-linearity (3 ReLU layers vs just 1)
→ The network can learn MORE complex features with FEWER parameters!

6.2 VGGNet Variants

Model	Layers	Parameters	Top-5 Error	Note
VGG-11	11	133M	10.9%	Lightest variant
VGG-16	16	138M	7.3%	Most widely used
VGG-19	19	144M	7.3%	Deepest variant

VGGNet (specifically VGG-16) became extremely popular and is still widely used today as a base model for transfer learning. Its simple, uniform structure makes it easy to understand and modify for custom tasks.

Limitation of VGGNet
VGGNet was very accurate but had a serious drawback:
• 138 million parameters → Requires HUGE amounts of memory (500MB+)
• Very slow to train — takes days even on powerful GPUs
• Three fully-connected layers at the end are very expensive

7. GoogLeNet / Inception (2014) Work Smarter, Not Just Deeper

7.1 Background

While VGGNet went wider and deeper, Google's team asked a completely different question: Instead of choosing which filter size to use, why not use ALL of them at once? This led to the birth of the Inception Module and GoogLeNet.

7.2 VGGNet vs GoogLeNet Side by Side

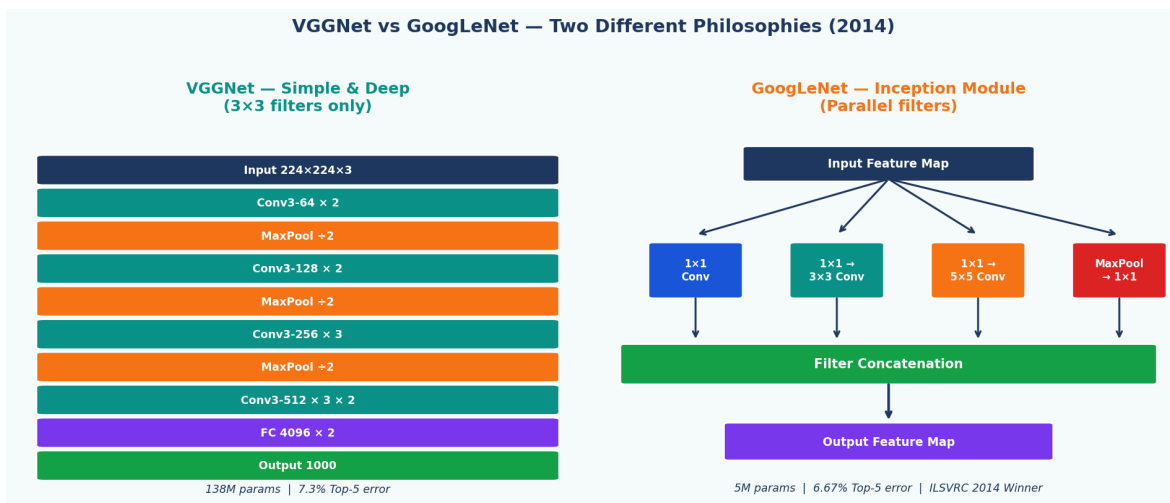


Figure 6: VGGNet (left) stacks uniform layers while GoogLeNet's Inception Module (right) runs multiple filter sizes in parallel

7.3 GoogLeNet Key Stats

- 22 layers deep much deeper than VGGNet
- Only 5 million parameters 27× fewer than AlexNet!
- Won ILSVRC 2014 with 6.67% Top-5 error rate
- Used Global Average Pooling instead of large Fully Connected layers
- Included auxiliary classifiers to fight the vanishing gradient problem

8. ResNet (2015) The Champion with Skip Connections

8.1 The Problem: Adding Layers Makes It Worse?

By 2015, researchers were trying to make CNNs even deeper 50, 100, even 1000 layers. But they found a shocking result: adding more layers actually made the network WORSE. A 56-layer network performed WORSE than a 20-layer one on training data!

The Degradation Problem

Expected: More layers = more learning = better accuracy

Reality: More layers = gradients vanish = network cannot learn effectively

If a 20-layer network works well, a 56-layer version should do at LEAST as well

— the extra 36 layers could simply learn the identity (copy input to output).

But they did NOT. Why? Because learning 'do nothing' is surprisingly HARD!

8.2 The Solution: Skip Connections (Residual Connections)

Kaiming He and his team at Microsoft Research had a brilliant idea: instead of learning the full transformation $H(x)$, let the network learn just the RESIDUAL the difference between what the output should be and the input.

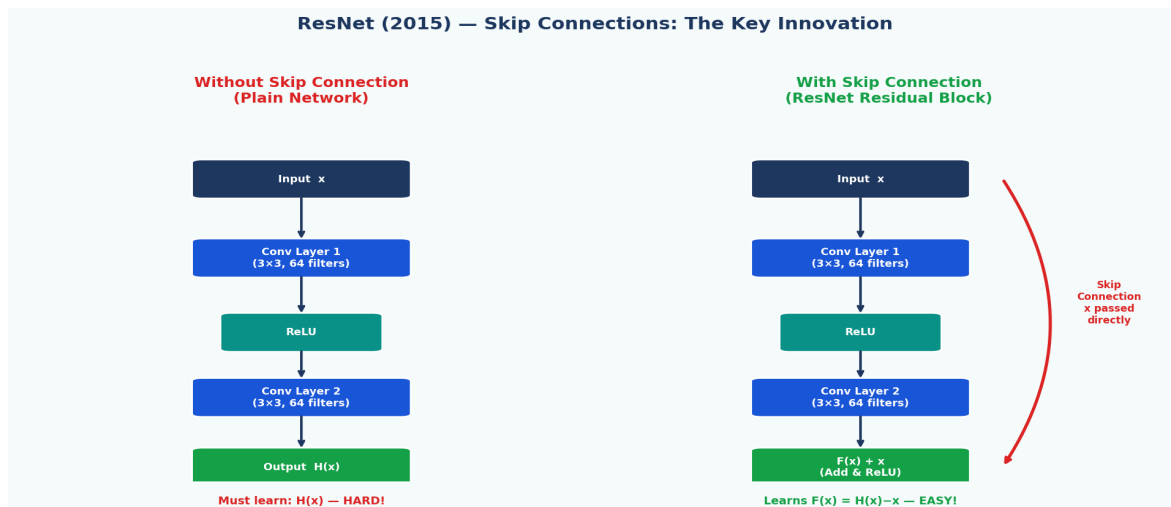


Figure 7: ResNet skip connection the shortcut lets the identity flow directly, so the network only needs to learn small corrections

8.3 Why Skip Connections Work

- If a layer is not needed, the network easily learns $F(x) = 0$, making output = x (identity)
- Gradients can flow directly through the skip connection, bypassing problematic layers
- This completely solves the vanishing gradient problem in very deep networks
- Training 100+ layer networks became stable, fast, and practical for the first time

8.4 ResNet Performance Results

Model	Layers	Parameters	Top-5 Error	Achievement
ResNet-18	18	11M	10.9%	Baseline
ResNet-34	34	21M	8.6%	Better than VGGNet
ResNet-50	50	25M	7.0%	Efficient deep model
ResNet-101	101	44M	6.4%	Near human level
ResNet-152	152	60M	5.71%	ILSVRC 2015 Winner

9. The Performance Race Visual Comparison

This chart shows the dramatic improvement in error rates across CNN architectures over time. Notice how ResNet-152 finally beats human-level performance (5.1% error):

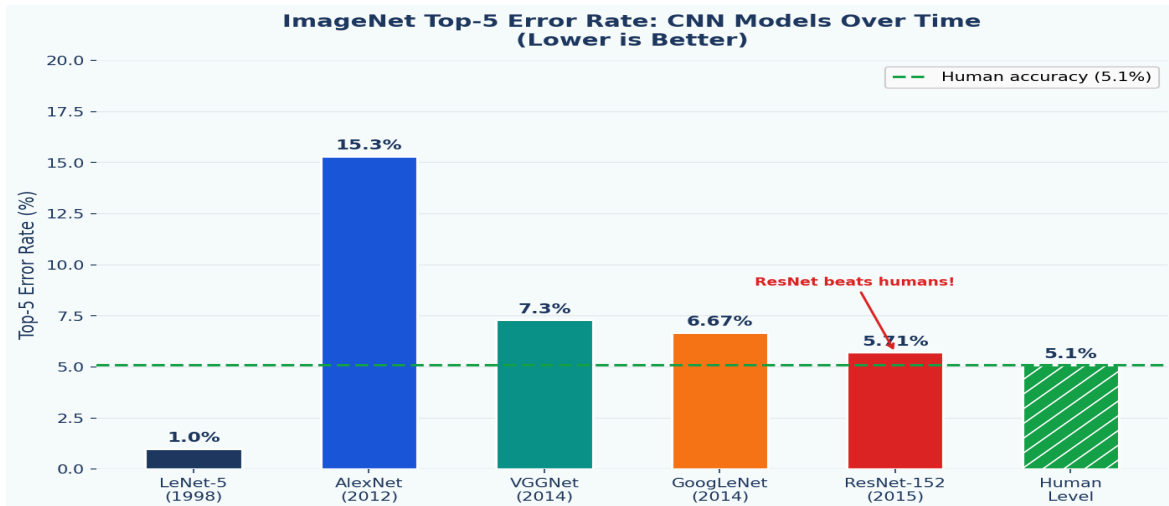


Figure 8: ImageNet Top-5 Error Rate comparison each architecture significantly improved over its predecessor

9.1 Master Comparison Table

Model	Year	Layer s	Params	Top-5 Error	Key Innovation
LeNet-5	1998	7	60K	~1% (MNIST)	First practical CNN
AlexNet	2012	8	60M	15.3%	ReLU, Dropout, GPU training
VGGNet-16	2014	16	138M	7.3%	Small 3×3 filters, uniform depth
GoogLeNet	2014	22	5M	6.67%	Inception module (parallel convs)
ResNet-152	2015	152	60M	5.71%	Skip connections (residuals)

10. Real-World Applications of CNNs Today

Understanding CNNs matters because they power countless things we use every day. Here are some exciting real-world applications:

Application	CNN Model Used	What It Does
Face Unlock on Phones	Custom CNN	Detects & recognises your face in milliseconds
Google Photos	ResNet	Groups similar faces, identifies objects in photos
Medical Diagnosis	ResNet-50	Detects cancer tumours in X-rays and MRI scans
Self-Driving Cars	VGGNet/YOLO	Identifies pedestrians, traffic signs, road lanes
Instagram Filters	CNN	Detects facial landmarks to apply AR effects
Satellite Imagery	ResNet	Maps land use, detects illegal deforestation
Manufacturing QA	AlexNet variants	Detects defects on production lines

11. Key Concepts Quick Reference Glossary

Essential Terms Explained Simply
CNN (Convolutional Neural Network): A neural network designed for processing image data
Convolution: A mathematical operation where a small filter slides over an image to detect features
Filter / Kernel: A small matrix (e.g., 3×3) that detects specific patterns like edges or textures
Feature Map: The output produced after applying a filter to an image
Pooling: Reducing the size of feature maps while keeping the most important information
ReLU: Rectified Linear Unit activation function defined as $f(x) = \max(0, x)$

Dropout: Randomly switching off some neurons during training to prevent overfitting
Vanishing Gradient: When gradients become too small to update early layers during training
Skip Connection: A shortcut that lets gradients flow directly across layers (ResNet's key idea)
Transfer Learning: Using a pre-trained CNN (like ResNet) as the starting point for a new task
Overfitting: When a model memorises training data but performs poorly on new unseen data

12. Summary

Let us recap the incredible journey of CNNs from 1998 to 2015:

1. LeNet (1998): The pioneer that proved CNNs could work for digit recognition only 60K params
2. AlexNet (2012): The breakthrough that launched the deep learning era with ReLU and GPU training
3. VGGNet (2014): Showed that simple, deep networks with small 3×3 filters achieve excellent results
4. GoogLeNet (2014): Proved efficiency matters Inception modules use multiple filter sizes in parallel
5. ResNet (2015): Solved the vanishing gradient problem with skip connections, enabling 150+ layer networks

Each architecture built upon the lessons of its predecessors. The story of CNNs is a beautiful example of scientific progress each generation of researchers asked 'what is the limitation of what we have?' and solved it creatively.

What Comes After ResNet?

The field did not stop at ResNet! Here are exciting developments after 2015:

- DenseNet (2016): Every layer connects to every other layer for maximum feature reuse
- MobileNet (2017): Tiny CNNs designed for phones and embedded/edge devices
- EfficientNet (2019): Automatically scales depth, width, and resolution together
- Vision Transformers (2020): Applies the Transformer (attention) architecture to images

The future of computer vision is still being written and YOU could be part of it!

13. References

- LeCun, Y. et al. (1998). Gradient-Based Learning Applied to Document Recognition. Proceedings of the IEEE, 86(11), 2278–2324.
- Krizhevsky, A., Sutskever, I., & Hinton, G. (2012). ImageNet Classification with Deep CNNs (AlexNet). NeurIPS.
- Simonyan, K., & Zisserman, A. (2014). Very Deep Convolutional Networks for Large-Scale Image Recognition (VGGNet). ICLR 2015.
- Szegedy, C. et al. (2014). Going Deeper with Convolutions (GoogLeNet/Inception). CVPR 2015.
- He, K. et al. (2015). Deep Residual Learning for Image Recognition (ResNet). CVPR 2016.
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep Learning. Nature, 521, 436–444.